Neutrino-Nucleon Structure Functions in Duality and DIS Regions

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Abstract

Purpose: Neutrino-nucleon charged-current deep inelastic scattering (DIS) in the transition region ($Q^2 \sim 1$–4 GeV^2) is critical for neutrino oscillation experiments (DUNE, Hyper-Kamiokande, NOvA, T2K), where perturbative QCD breaks down and non-perturbative hadronic effects become significant. This work constructs a hybrid description of weak structure functions F_2 and xF_3 across this crossover region, motivated by Bloom–Gilman quark–hadron duality. **Methods:** We combine the Regge-inspired CKMT parametrization at low Q^2 with the AMU perturbative QCD framework (NLO + target mass corrections + higher-twist effects) at higher Q^2. The models are matched at changeover scale $Q_0^2 = 2$ GeV^2 using two established transition prescriptions: a piecewise linear splice and a smooth sigmoid weighting function. In addition, the sensitivity to modern PDF sets (CT10nlo, CT18NLO, MMHT2014) is investigated for free nucleons. **Results:** Linear splicing of unscaled CKMT with AMU generates unphysical kink artefacts in xF_3 at low x due to curvature and normalization differences between the models. The smooth sigmoid weighting ($\Delta = 0.25$ GeV^2, AMU with LO+NLO+TMC+HT) restores derivative continuity across the matching boundary; however, a non-monotonic dip persists due to the normalization offset between unscaled CKMT and AMU, which is further deepened when higher-twist corrections are included. Predictions from different PDF sets show notable variation at high x and low Q^2, converging above the matching scale. **Conclusion:** The hybrid framework highlights the necessity of smooth matching and proper normalization across the resonance-to-DIS interface. Extension to nuclear targets and data comparison is ongoing.

Keywords: neutrino deep inelastic scattering; weak structure functions; CKMT parametrization; low-Q^2 modelling; quark–hadron duality; PDF sensitivity

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1 Introduction

Neutrino deep inelastic scattering (DIS) spans three kinematic regimes: quasi-elastic, resonance, and perturbative DIS. The few-GeV transition region ($Q^2 \sim 1$–4 GeV^2) is particularly challenging: significant event rates for experiments such as NOvA, T2K, DUNE, and Hyper-Kamiokande occur here, yet no single model covers it self-consistently [1].

Bloom–Gilman quark–hadron duality [2], and its extension to neutrino-induced weak processes [3], provides a theoretical bridge: resonance peaks, when plotted against the modified scaling variable $\omega' = (2M\nu + M^2)/Q^2$, oscillate around the DIS scaling curve and average to reproduce the DIS result. This motivates a systematic matching between low-Q^2 empirical parametrizations and perturbative QCD (pQCD) calculations.

Standard generators such as GENIE [9] and NuWro [10] rely on the Bodek–Yang (BY) prescription [4] to extend pQCD into the non-perturbative region, but BY is tied to GRV98 LO PDFs, preventing propagation of modern PDF uncertainties, and absorbs TMC and HT effects implicitly. In this work we construct a hybrid framework using the CKMT parametrization [5] at low Q^2 and an NLO+TMC+HT calculation (AMU model) at high Q^2, and compare two transition prescriptions for F_2 and xF_3 for free nucleons. Extension to nuclear targets is ongoing.

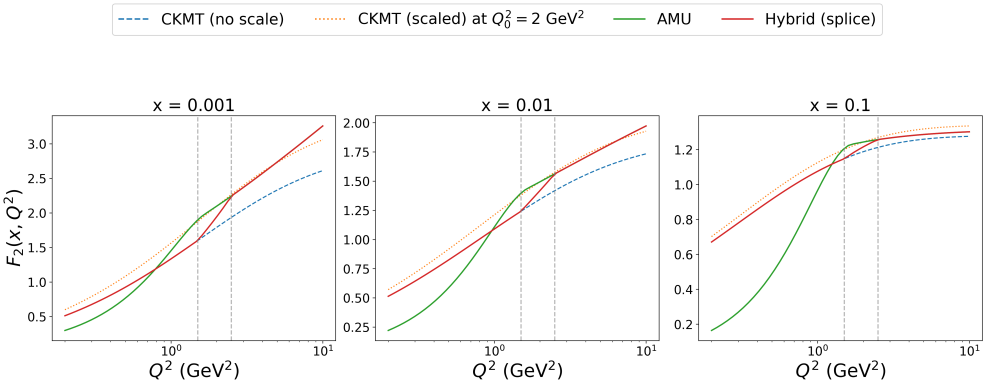


Figure 1: $F_2(x,Q^2)$ vs. Q^2 at $x = 0.001, 0.01, 0.1$ for unscaled CKMT (blue dashed), scaled CKMT at $Q_0^2 = 2$ GeV^2 (orange dotted), AMU (green solid), and linear splice (red solid). Vertical dashed lines mark the splice window $[1.5, 2.5]$ GeV^2.

2 Theoretical Framework

2.1 Charged-Current DIS Cross Section

The structure functions F_2 and xF_3 encode vector and parity-violating valence quark content respectively; xF_3 is particularly sensitive to the matching prescription since it probes valence quarks ($xF_3^{\nu N} = x[u_v + d_v]$ at leading order) [6]. For charged-current neutrino scattering $\nu_\mu(k) + N(p) \to \mu^-(k') + X(p')$, the double-differential cross section in the laboratory frame is [6]

$$\frac{d^2\sigma^{\nu,\bar{\nu}}}{dx\,dy} = \frac{G_F^2 M_N E_\nu}{\pi}\left(\frac{M_W^2}{Q^2+M_W^2}\right)^2\left[y^2 xF_1 + \left(1 - y - \frac{M_N x y}{2E_\nu}\right)F_2 \pm y\left(1-\frac{y}{2}\right)xF_3\right], \tag{1}$$

where $x = Q^2/(2M_N\nu)$ is the Bjorken variable, $y = \nu/E_\nu$ is the inelasticity, $Q^2 = -q^2$ is the momentum transfer squared, ν is the energy transfer in the lab frame, M_N is the nucleon mass, and G_F is the Fermi coupling constant. The upper (lower) sign $\pm$ in Eq. (1) applies to neutrino (anti-neutrino) scattering.

2.2 AMU (High-Q^2) Model

The AMU model computes the structure functions with CT10nlo PDFs [7], supplemented by:

- **NLO pQCD**: next-to-leading-order corrections via the DGLAP evolution equations.
- **TMC**: Target Mass Corrections following Schienbein et al. [8], important at moderate Q^2 and large x.
- **HT**: Higher Twist corrections ($\propto 1/Q^2$), suppressing structure functions at low Q^2 [11, 12].

The AMU model is reliable for $Q^2 \gtrsim 2$ GeV^2 but diverges in the non-perturbative regime.

2.3 CKMT Parametrization

The CKMT model [5] is a Regge-inspired empirical parametrization originally fitted to electromagnetic F_2 data, which is adapted to charged-current neutrino scattering for both $F_2^{\rm CC}$ and xF_3 following Jeong and Reno [6].

2.4 Hybrid Matching: Linear Splice

To bridge the low-Q^2 CKMT regime and high-Q^2 pQCD, the two models are matched across a crossover region around $Q_0^2 = 2.0$ GeV^2. While CKMT can be rescaled to enforce continuity at Q_0^2 ("scaled CKMT"), we preserve the original empirical parametrization ("unscaled CKMT") in our hybrid description. A piecewise linear interpolation splice across $[1.5, 2.5]$ GeV^2 is defined by:

$$F^{\rm hyb} = \begin{cases} F^{\rm CKMT} & Q^2 \le 1.5 \text{ GeV}^2, \\ c\,F^{\rm AMU} + (1-c)\,F^{\rm CKMT} & 1.5 < Q^2 < 2.5 \text{ GeV}^2, \\ F^{\rm AMU} & Q^2 \ge 2.5 \text{ GeV}^2, \end{cases} \tag{2}$$

with the linear weight $c(Q^2) = (Q^2 - 1.5)/(2.5 - 1.5)$.

2.5 Hybrid Matching: Smooth Sigmoid

Although the linear splice provides value continuity, differences in model curvature can induce derivative discontinuities and kink artefacts. To ensure a continuously differentiable transition, we replace the linear weight with a smooth sigmoid function, yielding the hybrid structure function:

$$F^{\rm hyb} = w(Q^2)\,F^{\rm CKMT} + \left[1 - w(Q^2)\right]F^{\rm AMU}, \tag{3}$$

with $w(Q^2) = \frac{1}{2}\left[1 - \tanh\left((Q^2 - Q_0^2)/\Delta\right)\right]$. We use a transition width $\Delta = 0.25$ GeV^2 centred at $Q_0^2 = 2.0$ GeV^2.

3 Results and Discussion

3.1 Hybrid Structure Functions: Linear Splice vs. Smooth Sigmoid

Figure 1 compares the original (unscaled) CKMT, scaled CKMT at $Q_0^2 = 2$ GeV^2, AMU, and the linear splice for F_2. For xF_3, because the linear splice directly joins the unscaled CKMT with AMU, the curvature and normalization mismatch generates a prominent downward kink and slope discontinuity at low x, visible in Fig. 2. Fig. 3 shows the result of replacing the linear weight with a smooth sigmoid ($\Delta = 0.25$ GeV^2). While the sigmoid restores derivative continuity and removes the sharp angular vertices, an unphysical non-monotonic dip remains visible across the transition window $[1.5, 2.5]$ GeV^2 because unscaled CKMT sits systematically above AMU at $Q^2 \sim 2$ GeV^2. Furthermore, HT corrections suppress AMU by 5–15% (purple dash-dotted vs. orange dashed), widening this normalization gap and deepening the dip in the full hybrid curve (green solid).

3.2 PDF Sensitivity

Figure 4 shows F_2 for three PDF sets (CT10nlo, CT18NLO, MMHT2014nlo68cl). At low x the sets converge before the splice window, leaving the hybrid effectively PDF-independent in that region. At high x (0.3, 0.6) the PDF spread is larger and extends closer to Q_0^2, motivating a full PDF uncertainty quantification.

4 Conclusions

We have presented a hybrid framework for the charged-current weak structure functions F_2 and xF_3 that bridges the non-perturbative CKMT description at low Q^2 and the AMU (NLO+TMC+HT) pQCD calculation at high Q^2:

1. The optimal changeover scale is $Q_0^2 = 2.0$ GeV^2.

2. Linear splicing produces visible kink artefacts for xF_3, most pronounced at low x.

3. Sigmoid weighting ($\Delta = 0.25$ GeV^2) restores derivative continuity, but an unphysical dip persists due to the normalization gap with unscaled CKMT, which is further accentuated by HT corrections.

4. PDF sensitivity is significant at high x and low Q^2, and vanishes above the matching scale.

Future work will include validation against CCFR/NuTeV structure-function data, extension to a wider range of nuclear targets, inclusion of PCAC/axial contributions, and a full PDF uncertainty quantification.

Conflict of Interest

The authors declare that they have no conflict of interest. No for-profit product is discussed or implied in this article.

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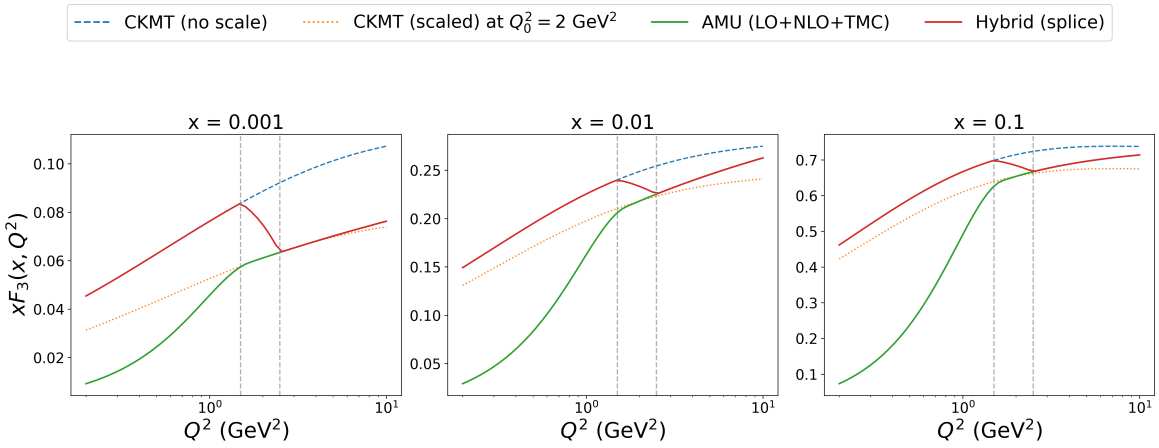


Figure 2: $xF_3(x, Q^2)$ vs. Q^2 for the linear splice: unscaled CKMT (blue dashed), scaled CKMT (orange dotted), AMU with TMC (green solid), and the hybrid (red solid). The kink and slope discontinuity are most prominent at low x.

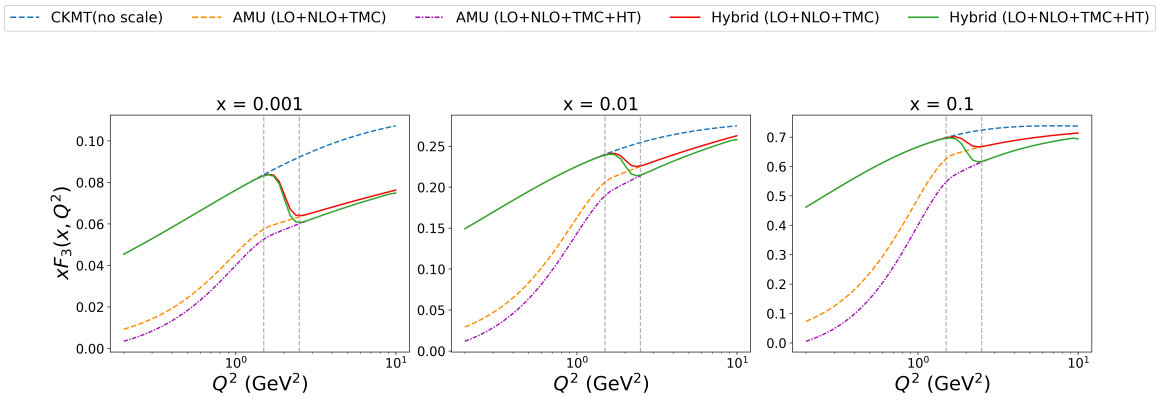


Figure 3: $xF_3(x, Q^2)$ vs. Q^2 with smooth sigmoid matching ($\Delta = 0.25$ GeV^2). Derivative continuity is restored, but the normalization offset leaves a non-monotonic dip; HT corrections deepen it slightly.

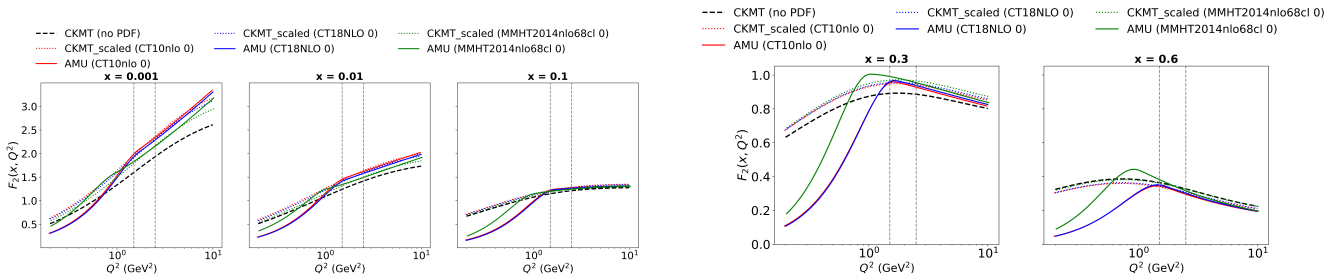


Figure 4: $F_2(x, Q^2)$ vs. Q^2 for CT10nlo, CT18NLO, and MMHT2014nlo68cl. The PDF spread is small at low x and larger at high x.